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(54) MEASURING DEVICE, CONTAINER MEANS AND SYSTEM FOR MONITORING AND MANAGING CONTAINER MEANS

MESSVORRICHTUNG, BEHÄLTERVORRICHTUNG UND SYSTEM ZUR ÜBERWACHUNG UND HANDHABUNG VON BEHÄLTERVORRICHTUNGEN

DISPOSITIF DE MESURE, MOYENS FORMANT CONTENANTS ET SYSTÈME PERMETTANT DE SURVEILLER ET DE GÉRER LES MOYENS FORMANT CONTENANTS

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Description

[0001] The invention relates to a measuring device, container means and a system for monitoring and managing container means that is suitable for containing various objects in multiple fields of application.

[0002] In particular, the measuring device according to the invention enables the filling status and/or the filling level of the container means to be measured in contexts in which the conferred objects are heterogeneous and unpredictable in terms of type, shape and material.

[0003] In particular, the container means is suitable for containing non-homogenous objects of different mature, for example unsorted waste, such as urban refuse.

[0004] US 2005/0217369 discloses the preamble of claim 1. CH557462 discloses an installation for collecting garbage and industrial waste into a container.

[0005] DE2504212 discloses an apparatus for disposing of residential, commercial and/or industrial waste.

[0006] US5465619 discloses an apparatus provided with a capacitive sensor for determining toner levels in an electrophotographic printing machine.

[0007] DE102007001175 discloses a capacitive device for the frill level measurement of corrosive media in a tank, particularly a urea solution for the after-treatment of waste gas in an internal combustion engine.

[0008] WO2008085060 discloses a method for measuring the liquid contents in a "bag in box" such as in wine boxes.

[0009] EP1818261 relates to a system and a method for the collection or transport of materials, such as trash.

[0010] WO2004110659 relates to a sclar powered compaction apparatus.

[0011] Skips with rather large dimensions are known for collecting urban waste, which are provided with sensors comprising load cells or strain-gauge elements that are suitable for measuring the weight of the refuse contained therein. Suck skips are able to send a signal indicating the weight contained therein. This signal can be received by an approaching vehicle designed to empty skips and collect waste.

[0012] Containers are further known for collecting waste that are provided with optic barrier sensors, or ultrasound sensors, which are suitable for supplying a signal when the filling level inside the containers reaches a set level. A system is provided for managing containers that provides for the aforesaid signal sent by the containers being received from a suitable station that plans the path of a collecting vehicle in function of the filling status of the various containers located in the urban area.

[0013] A drawback of the known containers discussed above is that they are rather complex to construct and entail rather a high cost. Further, maintenance and repair costs for possible damage to the aforesaid containers can be significant. From the economic point of view this makes it uneconomical to adopt such sensors on containers or at most makes it feasible to apply such sensors onto to containers of large dimensions in order to be able to amortise the cost thereof. Nevertheless, the containers of large dimensions are suitable for being distributed along roads and or/wide spaces but are not suitable for being located in zones such as pavements, parks, etc.

[0014] One object of the invention is to improve the measuring devices, the known containers, in particular containers for collecting waste, and the systems for managing the containers.

[0015] Another object of the invention is to provide a measuring device that is simple and economical in constructional, management and maintenance terms and is able to report the operating conditions or status of the container means and/or other information rapidly, precisely and efficiently.

[0016] The invention can be better understood and implemented with reference to the attached drawing that illustrate some embodiments thereof by way of non-limiting example, in which:

Figure 1 is a functional architecture diagram of container means provided with a measuring device and with a system for monitoring and managing this container means;

Figure 2 is a schematic section view of container means according to the invention;

Figure 3 shows schematically an operating mode of the system;

Figure 4 is another schematic view of operating modes of the system;

Figure 5 is a diagram illustrating various parts of the system including the container means according to the invention;

Figure 6 shows schematically various parts included in the container means;

Figure 7 is a diagram of a movable device included in the system;

Figure 8 is a diagram that shows how data and/or information are managed in the system according to the invention.

[0017] With reference to Figure 1, there is shown schematically container means 1 provided with a measuring device 10 according to the invention. The measuring device 10, which will be disclosed in detail further on, is used to measure and/or detect a filling status of the container means 1. In other words, the measuring device 10 acts to detect whether the container means 1 is in a total filling condition, or acts to detect possible occlusion at a mouth of the container means 1. Owing to a warning of the measuring device 10, it is possible in this manner to act promptly to empty the container means 1. The measuring device 10 can further be configured also for measuring the progressive filling of the container means 1, or for measuring periodically the filling level of the container means 1, as will be shown in detail by the description. Figure 1 shows schematically the functional architecture of a system 2 for monitoring and managing the container means 1. The system 2 can be used in various applica-

tions and/or industrial environments to detect and/or measure the degree of filling of container means 1 when the latter are intended to contain multiple objects that are heterogeneous and unpredictable in terms of type, shape and material. In other words, the container means 1 can receive non-homogenous objects of different nature.

[0018] In an application that is disclosed below in a non-limiting manner, the container means 1, or containers 1, are suitable for receiving and containing non-homogenous objects of various nature, for example unsorted waste, such as urban waste. In this case, a plurality of containers 1 is provided distributed over an urban area. The system 2 enables the filling status of the container means 1 to be monitored owing to the configuration of the latter that will be disclosed below, and enables waste collection and disposal operations to be managed in an optimum manner in an urban area.

[0019] The container means 1 is particularly suitable for being located at several points of the urban area, in particular in public gardens, on pavements, squares, etc. The container means 1 can have reduced dimensions so as to enable the positioning thereof on zones of any extent. In this case, the container means 1 is not bulky and does not necessarily require great space, unlike prior-art urban waste skips. The container means 1 is thus particularly suitable for receiving waste of various kinds from pedestrians and/or passers-by in any zone of the city. The container means 1 can have a height comprised between 40 cm and 140 cm, and can have a width and a depth comprised between 20 cm and 80 cm. In particular, the container means can have three different sizes. A first size, specified in Figure 1, provides a width and a depth that are substantially equal to 50 cm, and a height that is substantially equal to 80 cm. A second size provides a width and a depth that are substantially equal to 60 cm, and a height that is substantially equal to 100 cm. A third size provides a width and a depth that are substantially equal to 35 cm, and a height that is substantially equal to 50 cm. Nevertheless, the container means 1 with which to associate the measuring device 10 can have any desired dimension and shape.

[0020] The various parts that comprise the system 2, in particular the various parts of which the container means 1 is made up have great resistance to atmospheric agents and great mechanical resistance to blows, vibrations and wear.

[0021] The system 2 is configured for performing functions that according to requested requirements and to the degree of importance that it is desired to assign thereto comprise main functions F1, F2...Fn, which may be of basic importance, and secondary functions f1, f2...,fn, which may be optional and/or desired. The main functions that the system 2 can perform comprise: detecting the occlusion of the mouth of the container 1, detecting full or empty status of the container 1, temporal indication of the measurement. Further useful functions are: detecting the filling level of the container means 1, detecting a state of blockage or obstruction of the container means 1, localising the container means 1, promptly warning of the exceeding of the filling threshold of the container means 1. Information/data 26 acquired by the system 2 is sent to a departmental station 3 associated with a set territorial fraction of the town. The departmental station 3 comprises a data-gathering and analysis centre. In one embodiment, the departmental station 3 is configured for monitoring a plurality of containers 1 located in several zones of an area, such as an urban area, and further comprises a monitoring and management unit that is able to plan and programme the path of one or more vehicles employed to empty containers 1 and collect the waste contained therein. Several department stations 3 can be provided, each having distinct urban territorial competence, and a central station 19 to which all the departmental stations 3 are connected. Obviously, it is possible to have a single departmental station 3, which in this case coincides with the central station 19.

[0022] In Figure 2 there is shown the container means 1 according to the invention. The container means 1 comprises a casing 4 that defines a containing cavity 5 that is suitable for receiving objects, such as waste 6. The containing cavity 5 extends along a longitudinal axis A. The casing 4 is shaped so as to be able to house a bag 7 or similar elements that are suitable for receiving the waste 6 that users throw inside the container means 1. The bag 7 enables the container means 1 to be emptied rapidly, for example by an operator 12. The operator 12 extracts the bag 7 full of waste 6 from the container means 1 to transfer the waste to a collecting vehicle 13.

[0023] In the embodiment in Figure 2, the waste 6 is introduced into the container means 1 through an opening 8 associated with the casing 4.

[0024] The container means 1 comprises a lid element 9, which can be removably connected or can be hinged on the casing 4 and which is used to constrain an upper part of the bag 7 to the container means 1. In particular, the upper part of the bag 7, during use, is locked between the lid element 9 and an upper end zone of the casing 4.

[0025] The container means 1 is provided with the measuring device 10, configured for detecting the filling status of the containing cavity 5.

[0026] The device 10 is of capacitive type, i.e. it comprises one or more elements of capacitive type, i.e. one or more capacitors 14. The device 10 is provided with at least one pair of metal plates 11, each pair defining a capacitor 14. Each pair of plates 11 comprises a first pair 11a and a second pair 11b, arranged on or in the casing 4, on opposite sides to one another. In particular, the first metal plates 11a and the second metal plates 11b are positioned such as to be on opposite sides to the bag 7. In other words, the bag 7 is interposed between the first metal plates 11a and the second plates 11b. The set of plates 11 associated with the container means 1 defines a capacitor system 15.

[0027] The operating principle of the device 10 is based on the variation of the electric capacity of this capacitor system 15, through the effect of the introduction of one

or more objects inside the container means 1. In particular, the measuring device 10 detects variations in the value of electric capacity that are due to the introduction of objects, such as waste 6, inside the containing cavity 5.

[0028] When the containing cavity 5 is empty, or only the bag 7 is present, the electric capacity has a set value that substantially depends on the dielectric constant of the air. In this situation, it is possible to measure a capacity value of the capacitor system 15 "empty".

[0029] The various materials of which the waste consists that are deposited in the container means 1 are characterised by a dielectric constant that is different from the quality of the air.

[0030] By measuring a variation in the capacity of the capacitor system 15, it is possible to detect the presence of the waste.

[0031] An object is detected as introducing the object inside the containing cavity 5 determines a variation in the dielectric constant. Detecting an object can also occur through the effect of a variation in the geometry of the equivalent capacity. This occurs if the introduced object is of a conductive material that causes the resulting capacity to increase or decrease.

[0032] The measuring device 10 comprises an electronic unit 16 operationally connected to the plates 11, and provided with a local file 23, comprising a local memory 33. A supply source 17 is provided for supplying the electronic unit 16 and the plates 11. The supply source, in particular, comprises a battery 17. The battery 17 may comprise a rechargeable cell. A recharging device may be provided that is included in the container means 1, to recharge the battery 17. The recharging device can comprise a photovoltaic panel that is able to recharge the battery 17 with solar energy. In all cases, the type of power supply chosen is such as to have great resistance to environmental factors, require minimal maintenance, contain costs and simplify the management thereof and is such as to minimise environmental impact.

[0033] The device 10, owing to the electronic unit 16, is able to measure periodically and store the quantity of waste contained inside the container means 1.

[0034] By arranging several capacitors that are suitably positioned on the container means 1 it is possible to define filling "levels" and consequently to provide a discrete filling measurement.

[0035] As shown in the embodiment in Figure 2, the device 10 comprises a plurality of first metal plates 11a and of second plates 11b. Owing to this configuration, the device 10 is able to detect in a discrete and progressive manner various filling levels of the containing cavity 5.

[0036] The dimension and the position of the plates 11 is chosen in function of the degree of precision and discreteness of the measurement that it is desired to obtain. The greater the number of plates 11 and the less the dimension of the latter, the greater will be the resolution of the measurement and thus the greater will be the number of detectable and monitorable filling "levels". Each metal plate 11 can have a height comprised between 10 and 30 cm, and a width comprised between 30 and 70 cm. In the described embodiment, in particular but not in a limiting manner, each metal plate 11 has a height that is substantially equal to 21 cm, and a width that is substantially equal to 55 cm.

[0037] Owing to the measuring device 10 that is thus configured it is possible to obtain an effective and precise filling measurement of the container 1 that is independent of the type and/or geometry and/or materials of which the waste 6 is composed. The measuring device 10 is very reliable in reporting the filling level of the container 1 and prevents false negative signals on container 1 status from being generated.

[0038] With a suitable arrangement of several plates 11 and a processing algorithm of the signal/s supplied by the latter it is possible also to conduct shape measurements on the objects introduced into the containing cavity 5.

[0039] The measuring device 10 may also operate in continuous mode. This may be useful for counting the number of introductions of objects/waste into the container 1.

[0040] The measuring device 10 is also able to detect the occlusion of the containing cavity 5 by bags and other obstructive materials, regardless of whether the filling threshold has been reached. This occlusion measurement could occur through luminosity sensors, such as phototransistors or photocells suitably placed inside the containing cavity 5.

[0041] By adding further external sensors it is possible to count the number of persons in transit near the container 1.

[0042] The measuring device 10 is further configured to be mounted without problems on known containers that are already used for collecting waste. The measuring device 10 is able to be easily adapted to all known types and sizes of container having different geometrical shapes.

[0043] The measuring device 10 according to the invention is very cheap and also very difficult to tamper with and is therefore secure against acts of vandalism.

[0044] In a first embodiment, the container means 1 is configured for sending data directly to the departmental station 3 and/or to the central station 19 by communication means 24 of GPRS type included in the container means 1. In this case the container means 1 can be configured for transmitting data in real time. This can be provided for the entire area or for set zones of the latter. It may be useful to be able to dispose of data in real time in zones of particular/critical interest. It is possible to set up the container 1 in such a manner that the latter acquires data and information also from other similar containers of the zone and sends all the data to the central station 19 and/or to the departmental station 3. In this case a detailed analysis of a given set of containers 1 is conducted. At the end of this detailed analysis further containers 1 can be selected that are located in other

urban zones so as to define a new set of containers 1 to be monitored directly remotely in real time. The container 1 provided with the GPRS module could gather and convey the data of the containers placed near itself inside an area with a high density of containers (for example in which the distance between adjacent containers 1 is less than approximately 100 metres).

[0045] In a second embodiment, the system 2, with reference to Figure 3, comprises a movable device 18 configured for exchanging information with the container means 1, in particular for acquiring data from the latter. The movable device 18, which will be disclosed in detail in Figure 7, is provided for the operator 12, and/or can be provided on the collecting vehicle 13. In this last configuration, it is not necessary for the operator 12 to alight from the vehicle 13 to place the movable device 18 in contact with the container means 1. This is very useful for facilitating waste inspection and collection operations. The movable device 18 is configured for remote interaction with the container means 1. The movable device 18 communicates autonomously with the container means 1 when the operator 12 approaches the latter, thus acquiring information/data.

[0046] In particular, the data stored in the electronic unit 16 are sent to the movable device 18 to be subsequently transferred to the departmental station 3 and/or to the central operating station 19.

[0047] In particular, sending the data from the measuring device 10 to the movable device 18 can occur during the operations of replacing the bag 7 by the operator 12.

[0048] The container means 1 is identified univocally so as to be able to be discriminated on the basis of set historical analyses that take account of the intensity and the manner with which the container means 1 is used by the users.

[0049] At the moment of data collection the movable device 18 with which the operator 12 is provided also has to record the geographical position in which it has interacted with the container 1. This enables passive geolocation of all the containers 1, keeping the inventory up to date.

[0050] The operator 12, at the end of his shift, returns to the departmental station 3 and/or to the central station 19, where he can unload all the data contained in the movable device 18 through a wireless or cable connection.

[0051] At the central station 3 the data are stored in a general file 25 in order to be able to create the necessary reports and analyses. At the central station 3 the information/data 26 is processed to define analytical and synthetic representations 22 and is supplemented and correlated with cartographic data supplied by a cartographic database 20.

[0052] Figure 8 shows a mode with which the various information/data 26 relating to the containers 1 flows into the system 2. The data 26 saved on the movable devices 18 can be sent via Internet to an Internet server 27 where they are entered in the general file 25. In the general file 25 there is data identifying and locating the containers 1, data relating to current readings, and historical data associated with the various containers 1.

[0053] The information/data 26 is extracted from the general file 25 to be processed and integrated with information taken from the cartographic database 20, as indicated by the step F, so as to generate analytical and synthetic representations 22 that can be displayed graphically via a graphic user interface 28.

[0054] The departmental stations 3 and/or the central stations 19, owing to the user interface 28, are able to analyse and display various data, generate reports and documents and carry out or plan various intervention activities 29, for example maintenance activities, and generate alarm signals 30.

[0055] With reference to Figure 5 certain parts of the system 2 are disclosed, in particular the container 1 and the movable device 18 are disclosed in greater detail. In particular, in Figure 5 there is shown schematically the supply source 17 and the measuring device 10 comprising the metal plates 11 with which the container 1 and the electronic unit 16 are provided. The electronic unit 16 comprises a first processor 37 for processing data and/or signals by set algorithms. The container 1 further comprises a clock 31 and a temperature sensor 32.

[0056] The container 1 is provided with a local memory 33 that stores the information/data 26 that are subsequently sent to the departmental station 3 and/or to the central station 19 or to the movable device 18, with which the operator 12 is provided. The transmission of information/data 26 can take place via a first receiving-transmitting module 35 included in the container 1. The first receiving-transmitting module 35, depending on the transmission configurations that it is desired to obtain as disclosed previously, comprises a GPRS module and/or a short-range wireless module.

[0057] The container 1 is provided with a first interface 33 of serial and/or of JTAG type that conducts the diagnoses of various electronic parts of the container 1 rapidly and cheaply and programmes and/or debugs the various components of the electronic unit 16.

[0058] Figure 5 shows in greater detail the movable device 18 with which the operator 12 is provided. The movable device 18 comprises a second interface 36 of JTAG and/or serial and/or USB type, by means of which the movable device 18 is connectable to the container 1 and/or to the departmental station 3, and/or to the central station 19. The movable device 18 comprises a second receiving-transmitting module 40, for example of short-range wireless type, by means of which communication 39 occurs between the movable device 18 and the container 1.

[0059] The movable device 18 is provided with a second processor 38 for processing, using set algorithms, data and/or signals that are exchanged with the container 1 and/or with the departmental station 3, and/or with the central station 19. The movable device 18 can be con-

figured for being connected for example to the central station 19, in particular to the general file 25, by a wireless or USB connection so as to establish a communication 41 for exchanging and/or transferring data.

[0060] The movable device 18 is further provided with a supply battery 42 and with a local memory 43 for storing the information/data 26.

[0061] The movable device 18 comprises a GPS module 44 by means of which the position of the operator 12 and/or of each container 1 in the urban area is identified. The GPS module 44 is able to detect its own position at the moment of data gathering. In this manner, the GPS module 44 enables the position of the single container 1 at the moment of data collection to be ascertained indirectly.

[0062] The movable device 18 is provided with a user interface 45, comprising an acoustic warning horn, for example for the buzzer type, and/or a visual warning lamp, for example of the LED type. At the moment of data collection the movable device 18 further synchronises and/or corrects the clock 31 of the container 1.

[0063] With reference to Figure 6, there are shown in greater detail various electric/electronic parts included in the container 1, in particular in the electronic unit 16, some of which have already been disclosed with reference to Figure 5. An electric converter 46 is provided that acts on the electric current dispensed by the battery 17. The electric converter 46 can be configured for receiving a direct current input and for releasing a direct current output at a set voltage value. This value can, for example, be equal to 3.3 volt. The electronic unit 16 is provided with a microprocessor 47 suitable for processing and exchanging signals with other parts that are operationally connected thereto, for example with the local memory 33. A digital capacity meter 48 is provided for measuring the capacity value of the capacitor system 15, defined by the plates 11. A detecting device 49 is provided that is used to detect parameters of the electric current coming from the electric converter 46. The detecting device 49 is connected bidirectionally to the microprocessor 47 to send and/or receive information and/or command signals.

[0064] An antenna 50 is provided for sending information/data connected to the first receiving-transmitting module 35, already disclosed previously with reference to Figure 5. The electronic unit 16 is provided with a switch 51 connected operationally to the first receiving-transmitting module 35, to the digital meter 48, to the detecting device 49 and to the microprocessor 47.

[0065] The local memory 33, the digital capacity meter 48, the microprocessor 47, the first receiving-transmitting module 35, the antenna 50, the switch 51, the electric converter 46 and the detecting device 49 together define an electronic board 52.

[0066] The various parts disclosed above are connected together so as to permit operation in the desired modes. For example, the container 1 that is thus configured is able to store data relating to filling of the containing cavity 5 in given intervals of time that can be suitably set and selected. The container 1 that is thus configured, when it is interrogated by a movable device 18, is activated to send the data stored in the local memory 33. Alternatively and/or additionally, the container 1 can be configured for communicating autonomously with the departmental station 3 and/or with the central station 19, in particular for sending alarm signals relating to the filling status thereof, or for periodically sending various types of information the status or operating condition thereof.

[0067] With reference to Figure 7 below, various electric/electronic parts included in the movable device 18, are disclosed in greater detail, some of which have already been disclosed with reference to Figure 5.

[0068] The movable device 18 comprises a further electric converter 53 that acts on the electric current dispensed by the supply battery 42. The further electric converter 53 can be configured for receiving a direct current input and for releasing a direct current output at a set voltage value. This value can, for example, be equal to 3.3 volt. The further electric converter 53 is connected to the user interface 45.

[0069] A further detecting device 54 is provided that is used to detect parameters of the electric current coming from the further electric converter 53. The further detecting device 54 is connected bidirectionally to the second microprocessor 38 to send and/or receive information and/or command signals. The movable device 18 is provided with a USB port 55 by means of which the movable device 28 can be connected to an apparatus of the departmental station 3 and/or of the central station 19 to enable the information/data 26 to be exchanged. The memory 43, the GPS module 44, the USB port 55, the second receiving-transmitting module 40, to which an antenna 56 is connected, the further detecting device 54, the further electric converter 53, and the user interface 45 are operationally connected to the second processor 38. These parts together define a further electronic board 57. The measuring device 10 disclosed above is highly resistant to environmental factors and atmospheric agents, is reliable, has low consumption and requires no maintenance, or at most requires much reduced maintenance.

[0070] The system 2 is easy to manage and can be immediately understood by all operators, both those assigned to emptying the container means 1 and those working remotely in the departmental stations 3 and/or in the central station 19. The system 2 enables costs to be reduced because it optimises the waste collection cycle, being particularly advantageous, for example, for municipal administrations. The quality of the containers 1 emptying service is thus also improved, a cleaner town being thus obtained.

[0071] The solution proposed above for measuring filling status and, in particular, the filling status of container means, can be used not only in the field of managing waste but also in various other industrial applications and/or environments. The solution disclosed above can

be adapted to containers having any desired geometrical shape and dimensions.

[0072] Variations and/or additions to what have been disclosed above and illustrated in the attached drawings are possible.

Claims

1. A container comprising a casing (4) defining a containing cavity (5) for receiving non-homogenous objects and including a measuring device for detecting a filling status of said containing cavity (5), comprising capacitive means (11, 11a, 11b, 14, 15) that is positionable near said containing cavity (5) and configured for detecting a variation in electric capacity caused by the presence of said objects (6) so as to determine said filling status, wherein said capacitive means comprise plate means (11, 11a, 11b) that is positionable in peripheral zones of said containing cavity (5), configured for detecting a total filling status or a partial filling status of said containing cavity (5), said capacitive means comprising first plate means (11a) and second plate means (11b), mutually arranged at opposite ends with respect to said containing cavity (5) and arranged on or in said casing (4), said first plate means and said second plate means comprising respectively a plurality of first plate elements (11a) and a plurality of second plate elements (11b) which are positionable in a manner distributed parallel to a longitudinal axis (A) of said containing cavity (5) so as to be able to detect a plurality of progressive filling levels of said containing cavity (5), the introduction of an object inside said containing cavity (5) determining a variation in the dielectric constant or in the geometry of the equivalent capacity, **characterised in that** said container is a waste container (1) for containing said non-homogenous objects, said casing (4) being shaped so as to be able to house a bag (7) or similar elements for receiving said non-homogenous objects, said bag (7) or similar elements being intended to be interposed between said plurality of first plate elements (11a) and said plurality of second plate elements (11b).

2. A container according to claim 1, wherein said first plate means (11a) and said second plate means (11b) lie on surfaces that laterally bound said containing cavity (5) and which are arranged transversely to a plane defined by an opening (8) through which said non-homogenous objects are introduced into said containing cavity (5).

3. A container according to claim 1 or 2, wherein each of said first plate elements (11a) and of said second plate elements (11b) has a height comprised between 10 and 30 cm, in particular approximately equal to 21 cm, and a width comprised between 30 and 70 cm, in particular approximately equal to 55 cm.

4. A container according to any preceding claim, further comprising an electronic unit (16) provided with a processor (37) for processing data and/or signals and a digital capacity meter (48).

5. A container according to claim 4 and further comprising a local memory (33) for storing information/data (26), supplied in particular by said processor (37).

6. A container according to any preceding claim, further comprising a first receiving-transmitting module (35) for transmitting data.

7. A container to claim 6, wherein said first receiving-transmitting module (35) comprises a GPRS module or is configured for short-range wireless transmission.

8. A container according to any preceding claim, further comprising a supply source (17) by means of which said measuring device (10) is electrically suppliable.

9. A container according to claim 8, wherein said supply source comprises a battery (17), in particular a battery that is rechargeable by recharging photovoltaic means.

10. A container according to any preceding claim, further comprising an interface (33) of serial and/or of JTAG type.

11. A container according to any preceding claim, further comprising a clock (31) and a temperature sensor (32).

12. A container according to any preceding claim, further comprising luminosity sensor means for detecting an occlusion of said containing cavity (5) - caused, for example, by bags or other bulky objects - independently of whether a filling threshold of said containing cavity (5) has been reached.

13. A container according to claim 12, wherein said luminosity sensor means, which is positionable in said containing cavity (5), is chosen from a group comprising: phototransistor sensors, photocell sensors.

14. A container according to claim 13, and having a height comprised between 40 cm and 140 cm, in particular approximately equal to 80 cm, and a width and a depth comprised between 20 cm and 80 cm, in particular approximately equal to 50 cm.

15. System for monitoring and managing urban waste comprising a container according to any one of claims 1 to 14 and a station (3; 19) configured for receiving and processing data provided by said measuring device (10) so as to monitor said filling status and plan intervention activities on and/or manage, said containing cavity (5).

16. system according to claim 15, wherein said station (3; 19) is configured for receiving data sent by said measuring device (10) via a wireless connection of GPRS type.

17. System according to claims 15 or 16, and further comprising a movable device (18) for acquiring data from said measuring device (10) to transfer the data to said station (3; 19).

18. System according to claim 17, wherein said movable device (18) is configured for being connected to said measuring device (10) via a wireless connection.

19. System according to claim 17 or 18, wherein said movable device (18) is provided with a memory (43) and with a further interface (36) of JTAG and/or serial and/or USB type by means of which said movable device (18) is connectable to said measuring device (10) and/or to said station (3; 19).

20. System according to any one of claims 17 to 19, wherein said movable device (18) comprises a GPS module for geolocating said containing cavity (5).

21. System according to any one of claims 17 to 20, wherein said movable device (18) is provided with acoustic-visual warning means (45).

Patentansprüche

1. Behälter mit einem Gehäuse (4), das einen Aufnahmehohlraum (5) definiert, um nicht homogene Objekte aufzunehmen, und mit einer Messeinrichtung zum Detektieren eines Füllstatus des Aufnahmehohlraums (5), mit einem Kapazitätsmittel (11, 11a, 11 b, 14, 15), das in der Nähe des Aufnahmehohlraums (5) positionierbar ist und ausgebildet ist, um eine Veränderung der elektrischen Kapazität, verursacht durch das Vorhandensein der genannten Objekte (6) zu detektieren, um so den Füllstatus zu bestimmen, wobei das Kapazitätsmittel ein Plattenmittel (11, 11 a, 11 b) aufweist, das in Umfangszonen des genannten Hohlraums (5) positionierbar ist und dazu ausgebildet ist, einen Gesamtfüllstatus oder einen Teilfüllstatus des Aufnahmehohlraums (5) zu detektieren, wobei das Kapazitätsmittel ein erstes Plattenmittel (11a) und ein zweites Plattenmittel (11 b) aufweist, die gemeinsam an gegenüberliegenden Enden in Bezug auf den Aufnahmehohlraum (5) positioniert sind und auf oder in dem Gehäuse (4) angeordnet sind, wobei das erste Plattenmittel und das zweite Plattenmittel jeweils eine Mehrzahl von ersten Plattenelementen (11a) und Mehrzahl von zweiten Plattenelementen (11 b) aufweist, die in einer Weise positionierbar sind, die parallel zu einer Längsachse (A) des Aufnahmehohlraums (5) verteilt sind, um so in der Lage zu sein, eine Mehrzahl von progressiven Füllpegeln des Aufnahmehohlraums (5) zu detektieren, wobei das Einführen eines Objektes in den Aufnahmehohlraum (5) eine Veränderung in der dielektrischen Konstante oder in der Geometrie der äquivalenten Kapazität bewirkt, **dadurch gekennzeichnet, dass** der Behälter ein Abfallbehälter (1) ist, um die nicht homogenen Objekte zu halten, wobei das Gehäuse (4) dazu geformt ist, eine Tasche (7) oder ähnliche Elemente zum Aufnehmen der nicht homogenen Objekte zu halten, wobei die Tasche (7) oder die ähnlichen Elemente zwischen der Mehrzahl der ersten Plattenelementen (11a) und der Mehrzahl der zweiten Plattenelementen (11 b) aufgenommen sein sollen.

2. Behälter nach Anspruch 1, bei dem das erste Plattenmittel (11a) und das zweite Plattenmittel (11 b) auf Flächen liegen, die den Aufnahmehohlraum (5) seitlich einschließen und die quer zu einer Ebene angeordnet sind, die durch eine Öffnung (8), durch die die nicht homogenen Objekte in den Aufnahmehohlraum (5) eingeführt werden, definiert ist.

3. Behälter nach Anspruch 1 oder 2, bei dem jedes der ersten Plattenelemente (11a) und der zweiten Plattenelemente (11 b) eine Höhe aufweist, die zwischen 10 und 30 cm liegt, insbesondere ungefähr gleich 21 cm ist, und eine Breite, die zwischen 30 und 70 cm liegt, insbesondere ungefähr gleich 55 cm ist.

4. Behälter nach irgendeinem vorhergehenden Anspruch, der ferner eine Elektronikeinheit (16) aufweist, die mit einem Prozessor (37) versehen ist, um Daten und/oder Signale zu verarbeiten, sowie ein digitales Kapazitätsmessgerät (48).

5. Behälter nach Anspruch 4, der ferner einen lokalen Speicher (33) aufweist, um Information/Daten (26) zu speichern, die insbesondere von dem Prozessor (37) geliefert werden.

6. Behälter nach irgendeinem vorhergehenden Anspruch, der ferner ein erstes Empfänger-/Sender-Modul (35) zum Übertragen von Daten aufweist.

7. Behälter nach Anspruch 6, bei dem das erste Empfänger-/Sender-Modul (35) ein GPRS-Modul aufweist oder zur drahtlosen Übertragung im Nahbereich ausgebildet ist.

8. Behälter nach irgendeinem vorhergehenden Anspruch, der ferner eine Versorgungsquelle (17) aufweist, wodurch die Messvorrichtung (10) mit Elektrizität versorgt wird.

9. Behälter nach Anspruch 8, bei dem die Versorgungsquelle eine Batterie (17) aufweist, insbesondere eine Batterie, die mittels eines Photovoltaikmittels wieder aufladbar ist.

10. Behälter nach irgendeinem vorhergehenden Anspruch, der ferner eine Schnittstelle (33) der seriellen Art und/oder der JTAG-Art aufweist.

11. Behälter nach irgendeinem vorhergehenden Anspruch, der ferner eine Uhr (31) und einen Temperatursensor (32) aufweist.

12. Behälter nach irgendeinem vorhergehenden Anspruch, der ferner ein Helligkeitssensormittel zum Detektieren einer Okklusion des Aufnahmehohlraums (5) aufweist, die zum Beispiel durch Taschen oder andere ausladende Objekte bedingt ist, unabhängig davon, ob eine Füllgrenze des Aufnahmehohlraums (5) erreicht wurde.

13. Behälter nach Anspruch 12, bei dem das Helligkeitssensormittel, das in dem Aufnahmehohlraum (5) positionierbar ist, aus einer Gruppe ausgewählt ist, die besteht aus: Fototransistorsensoren, Fotozellensensoren.

14. Behälter nach Anspruch 13, der eine Höhe aufweist, die zwischen 40 cm und 140 cm liegt, insbesondere ungefähr gleich 80 cm ist, und eine Breite und eine Tiefe, die zwischen 20 cm und 80 cm liegt, insbesondere ungefähr gleich 50 cm ist.

15. System zum Überwachen und Managen von kommunalem Müll, das einen Behälter gemäß irgendeinem der Ansprüche 1 bis 14 und eine Station (3; 19) aufweist, die zur Aufnahme und Verarbeitung von Daten ausgebildet ist, die von der Messvorrichtung (10) geliefert werden, um so den Füllstatus und Planungseingriffsaktivitäten betreffend den Aufnahmehohlraum (5) zu überwachen und/oder zu managen.

16. System nach Anspruch 15, bei dem die Station (3; 19) dazu ausgebildet ist, Daten zu erhalten, die von der Messvorrichtung (10) über eine drahtlose Verbindung der GPRS-Art gesendet werden.

17. System nach Anspruch 15 oder 16, das ferner eine bewegliche Vorrichtung (18) zum Erhalten von Daten von der Messvorrichtung (10) aufweist, um die Daten zu der Station (3; 19) zu übertragen.

18. System nach Anspruch 17, bei dem die bewegliche Vorrichtung (18) dazu ausgebildet ist, mit der Messvorrichtung (10) über eine drahtlose Verbindung verbunden zu werden.

19. System nach Anspruch 17 oder 18, bei dem die bewegliche Vorrichtung (18) mit einem Speicher (43) versehen ist, sowie mit einer weiteren Schnittstelle (36) der JTAG-Art und/oder der seriellen Art und/oder der USB-Art, wodurch die bewegliche Vorrichtung (18) mit der Messvorrichtung (10) und/oder mit der Station (3; 19) verbindbar ist.

20. System nach irgendeinem der Ansprüche 17 bis 19, bei dem die bewegliche Vorrichtung (18) ein GPS-Modul zur Lageerfassung des Aufnahmehohlraums (5) aufweist.

21. System nach irgendeinem der Ansprüche 17 bis 20, bei dem die bewegliche Vorrichtung (18) mit akustischen/visuellen Warnmitteln (45) versehen ist.

Revendications

1. Conteneur comprenant un boîtier (4) qui définit une cavité de confinement (5) destinée à recevoir des objets non homogènes et qui inclut un dispositif de mesure pour la détection d'un état de remplissage de ladite cavité de confinement (5), comprenant des moyens capacitifs (11, 11a, 11b, 14, 15) qui peuvent être positionnés au voisinage de ladite cavité de confinement (5) et sont configurés pour détecter une variation de capacité électrique provoquée par la présence desdits objets (6) afin de déterminer ledit état de remplissage, dans lequel lesdits moyens capacitifs comprennent des moyens de plaque (11, 11a, 11b) qui peuvent être positionnés dans des zones périphériques de ladite cavité de confinement (5) et sont configurés pour détecter un état de remplissage total ou un état de remplissage partiel de ladite cavité de confinement (5), lesdits moyens capacitifs comprenant des premiers moyens de plaque (11a) et des deuxièmes moyens de plaque (11b), disposés mutuellement à des extrémités opposées par rapport à ladite cavité de confinement (5) et disposés sur ou dans ledit boîtier (4), lesdits premiers moyens de plaque et lesdits deuxièmes moyens de plaque comprenant respectivement une pluralité de premiers éléments de plaque (11a) et une pluralité de deuxièmes éléments de plaque (11b) qui peuvent être positionnés de manière à être répartis parallèlement à un axe longitudinal (A) de ladite cavité de confinement (5) afin d'être aptes à détecter une pluralité de niveaux de remplissage progressifs de ladite cavité de confinement (5), l'introduction d'un objet à l'intérieur de ladite cavité de confinement (5) déterminant une variation de la constante diélectrique ou de la géométrie de la capacité équivalente,

caractérisé en ce que ledit conteneur est un conteneur de déchets (1) destiné à contenir des objets non homogènes, ledit boîtier (4) étant conformé de manière à être apte à loger un sac (7) ou des éléments similaires pour recevoir lesdits objets non homogènes, ledit sac (7) ou lesdits éléments similaires étant destiné(s) à être interposé(s) entre ladite pluralité des premiers éléments de plaque (11a) et ladite pluralité des deuxièmes éléments de plaque (11b).

2. Conteneur selon la revendication 1, dans lequel lesdits premiers éléments de plaque (11a) et lesdits deuxièmes éléments de plaque (11b) sont situés sur des surfaces qui limitent latéralement ladite cavité de confinement (5) et qui sont disposées transversalement par rapport à un plan défini par une ouverture (8) à travers laquelle lesdits objets non homogènes sont introduits dans ladite cavité de confinement (5).

3. Conteneur selon la revendication 1 ou 2, dans lequel chacun desdits premiers éléments de plaque (11a) et desdits deuxièmes éléments de plaque (11b) a une hauteur comprise entre 10 et 30 cm, plus particulièrement égale approximativement à 21 cm, et une largeur comprise entre 30 et 70 cm, plus particulièrement égale approximativement à 55 cm.

4. Conteneur selon une quelconque revendication précédente, comprenant en outre une unité électronique (16) munie d'un processeur (37), destiné à traiter des données et/ou des signaux, et d'un dispositif de mesure de capacité numérique (48).

5. Conteneur selon la revendication 4 et comprenant en outre une mémoire locale (33) destinée à stocker des informations/des données (26), fournies en particulier par ledit processeur (37).

6. Conteneur selon une quelconque revendication précédente, comprenant en outre un premier module d'émission-réception (35) destiné à la transmission de données.

7. Conteneur selon la revendication 6, dans lequel ledit premier module d'émission-réception (35) comprend un module GPRS ou est configuré pour une transmission sans fil à courte distance.

8. Conteneur selon une quelconque revendication précédente, comprenant en outre une source d'alimentation (17) au moyen de laquelle ledit dispositif de mesure (10) peut être alimenté électriquement.

9. Conteneur selon la revendication 8, dans lequel ladite source d'alimentation comprend une batterie (17), en particulier une batterie qui est rechargeable à l'aide de moyens de recharge photovoltaïques.

10. Conteneur selon une quelconque revendication précédente, comprenant en outre une interface (33) de type série et/ou de type JTAG.

11. Conteneur selon une quelconque revendication précédente, comprenant en outre une horloge (31) et un capteur de température (32).

12. Conteneur selon une quelconque revendication précédente, comprenant en outre des moyens de détection de luminosité destinés à détecter une occlusion de ladite cavité de confinement (5) - provoquée, par exemple, par des sacs ou d'autres objets volumineux - indépendamment du fait de savoir si un seuil de remplissage de ladite cavité de confinement (5) a été atteint.

13. Conteneur selon la revendication 12, dans lequel lesdits moyens de détection de luminosité, qui peuvent être positionnés dans ladite cavité de confinement (5), sont choisis dans un groupe comprenant : des capteurs à phototransistors, des capteurs à cellules photoélectriques.

14. Conteneur selon la revendication 13, et ayant une hauteur comprise entre 40 et 140 cm, plus particulièrement égale approximativement à 80 cm, et une largeur et une profondeur comprises entre 20 et 80 cm, plus particulièrement égale approximativement à 50 cm.

15. Système de surveillance et de gestion de déchets urbains comprenant un conteneur selon l'une quelconque des revendications 1 à 14 et une station (3 ; 19) configurée pour recevoir et traiter des données fournies par ledit dispositif de mesure (10) de manière à surveiller ledit état de remplissage et planifier des activités d'intervention sur ladite cavité de confinement (5) et/ou gérer celle-ci.

16. Système selon la revendication 15, dans lequel ladite station (3 ; 19) est configurée pour recevoir des données envoyées par ledit dispositif de mesure (10) par l'intermédiaire d'une connexion sans fil de type GPRS.

17. Système selon les revendications 15 ou 16, et comprenant en outre un dispositif mobile (18) pour l'acquisition de données provenant dudit dispositif de mesure (10) afin de transférer les données à ladite station (3 ; 19).

18. Système selon la revendication 17, dans lequel ledit dispositif mobile (18) est configuré pour être connecté audit dispositif de mesure (10) par l'intermédiaire d'une connexion sans fil.

19. Système selon la revendication 17 ou 18, dans le-

quel ledit dispositif mobile (18) est muni d'une mémoire (43) et d'une interface supplémentaire (36) de type JTAG et/ou de type série et/ou de type USB au moyen desquelles ledit dispositif mobile (18) peut être connecté audit dispositif de mesure (10) et/ou à ladite station (3 ; 19).

20. Système selon l'une quelconque des revendications 17 à 19, dans lequel ledit dispositif mobile (18) comprend un module GPS de géolocalisation de ladite cavité de confinement (5).

21. Système selon l'une quelconque des revendications 17 à 20, dans lequel ledit dispositif mobile (18) est muni de moyens d'avertissement sonores et visuels (45).

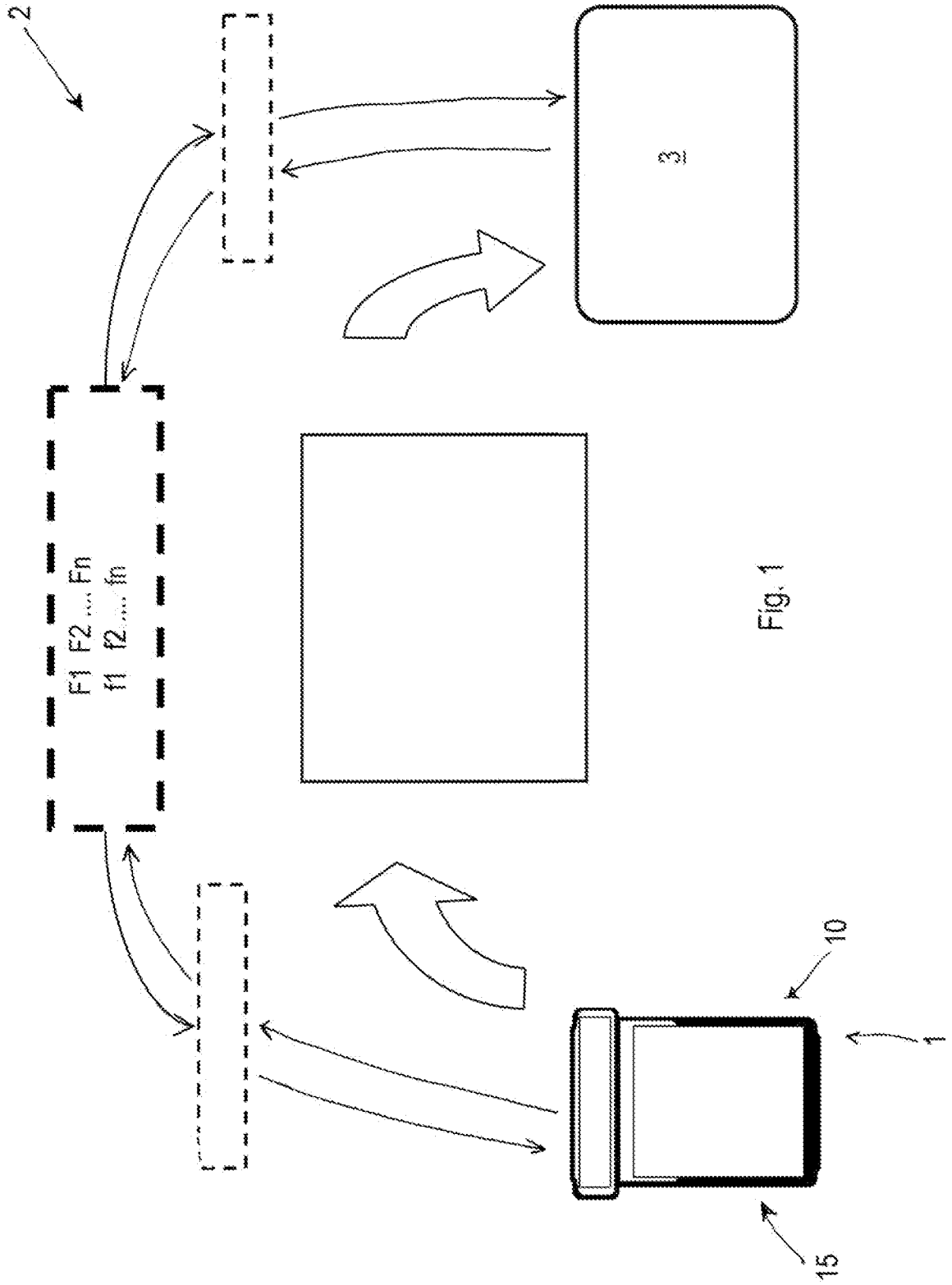


Fig. 1

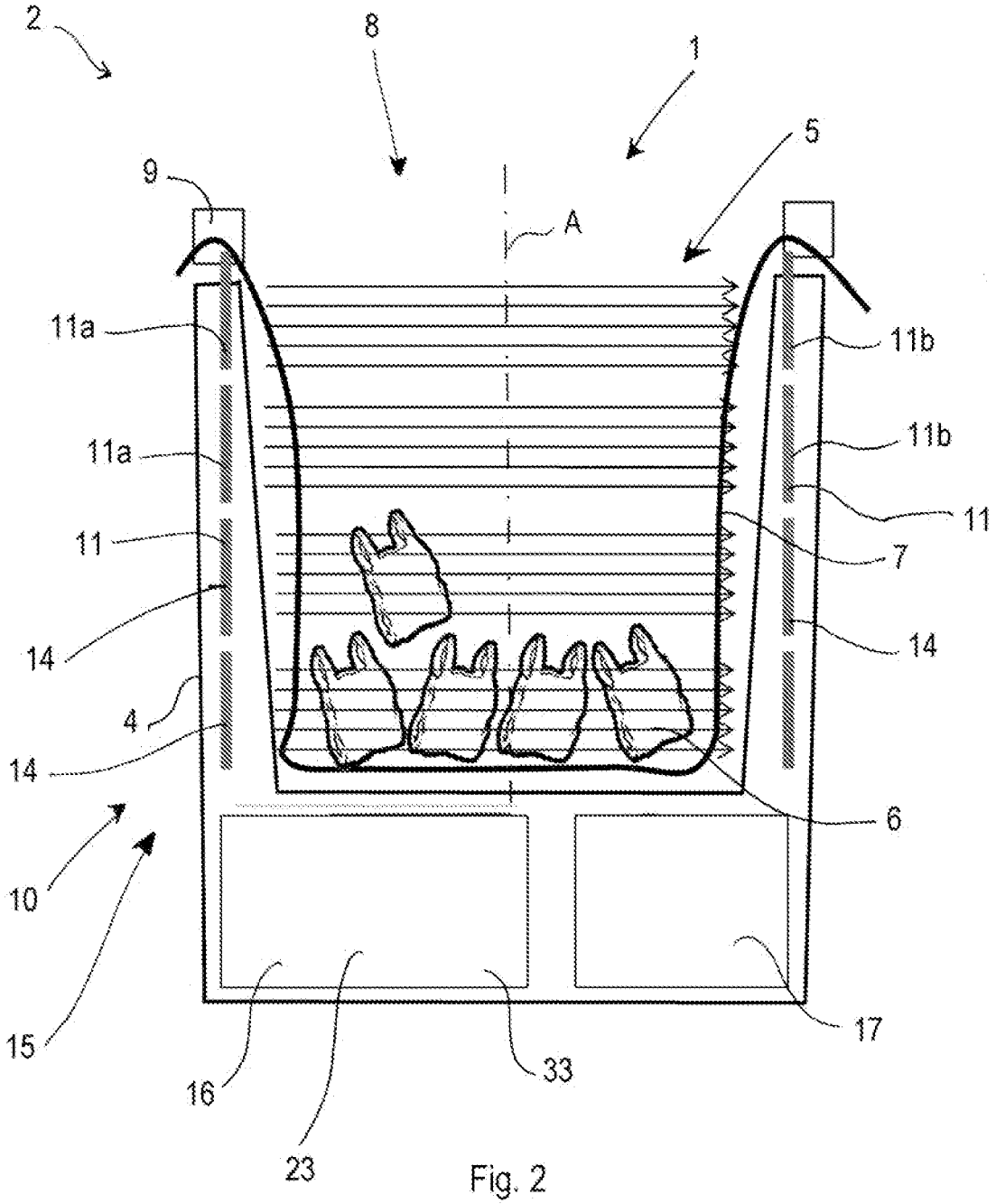


Fig. 2

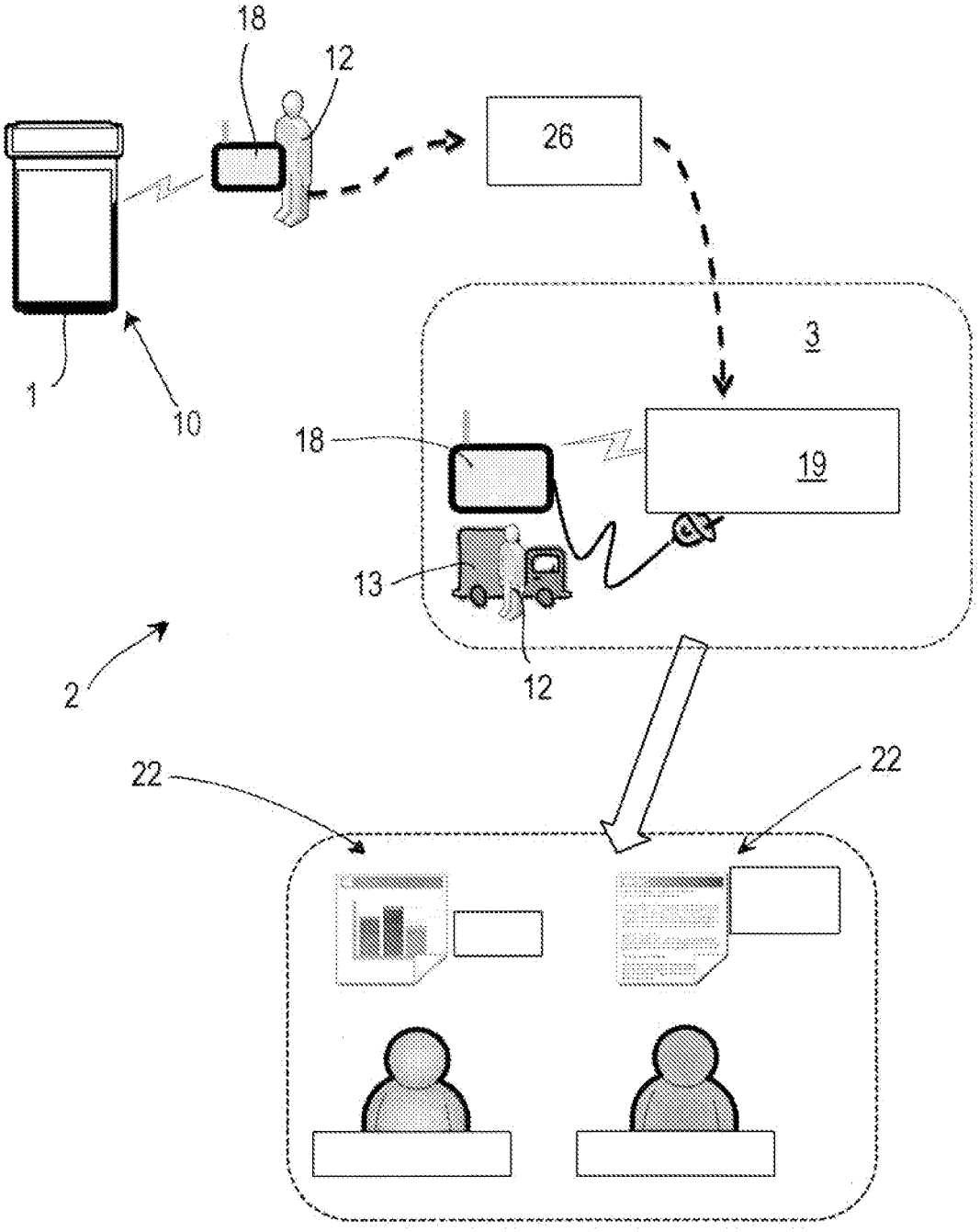


Fig. 3

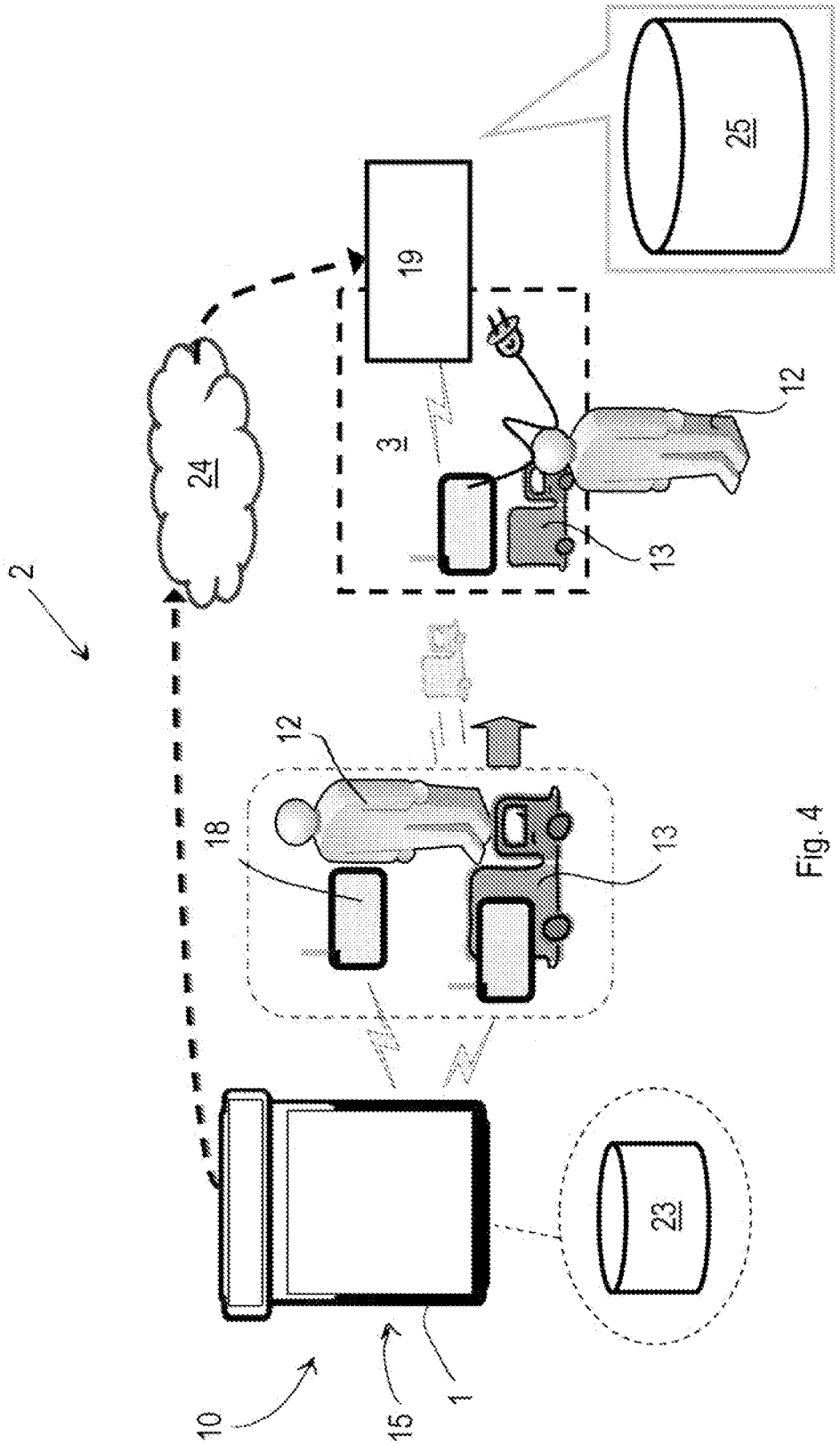


Fig. 4

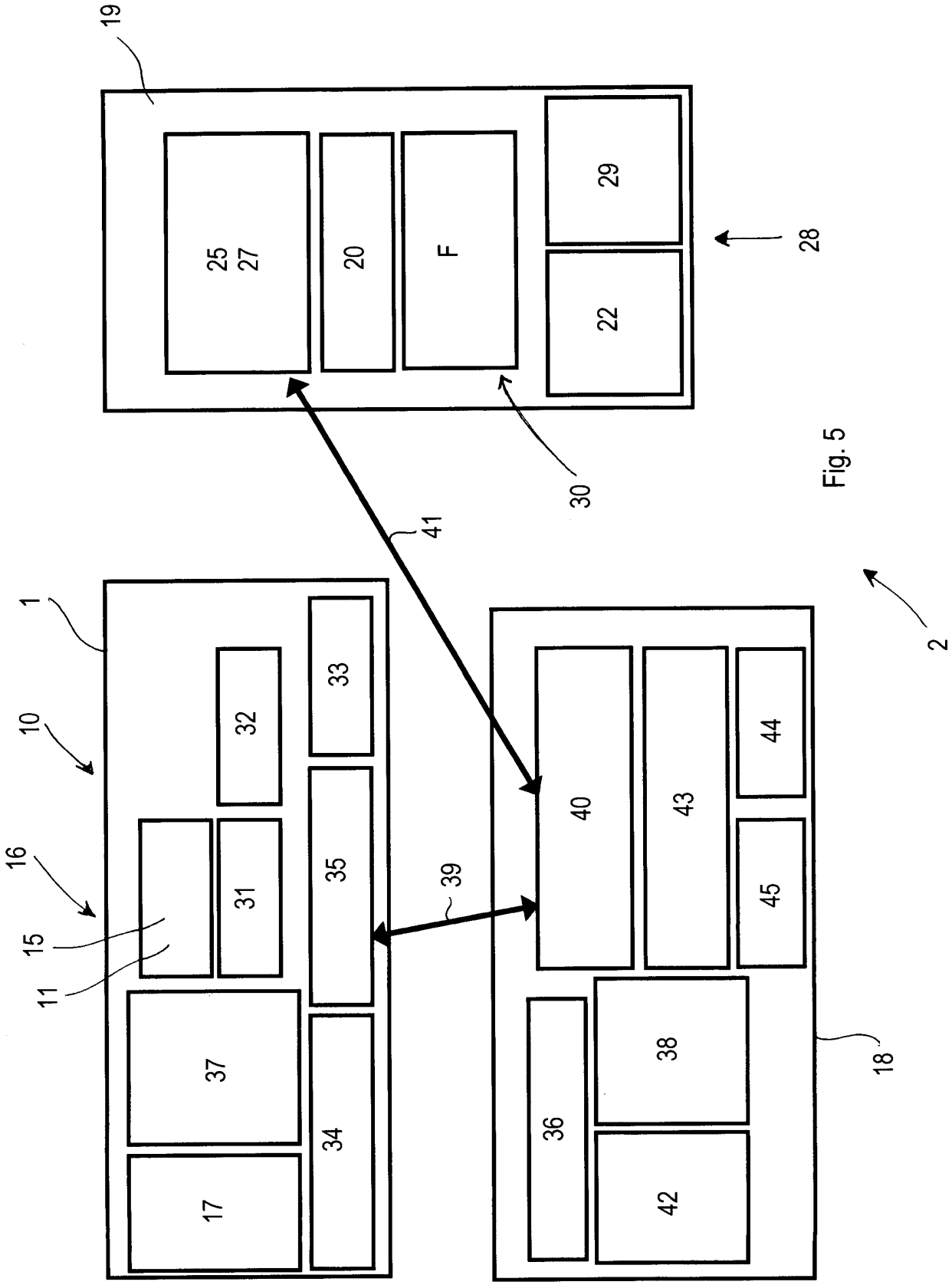

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Fig. 5

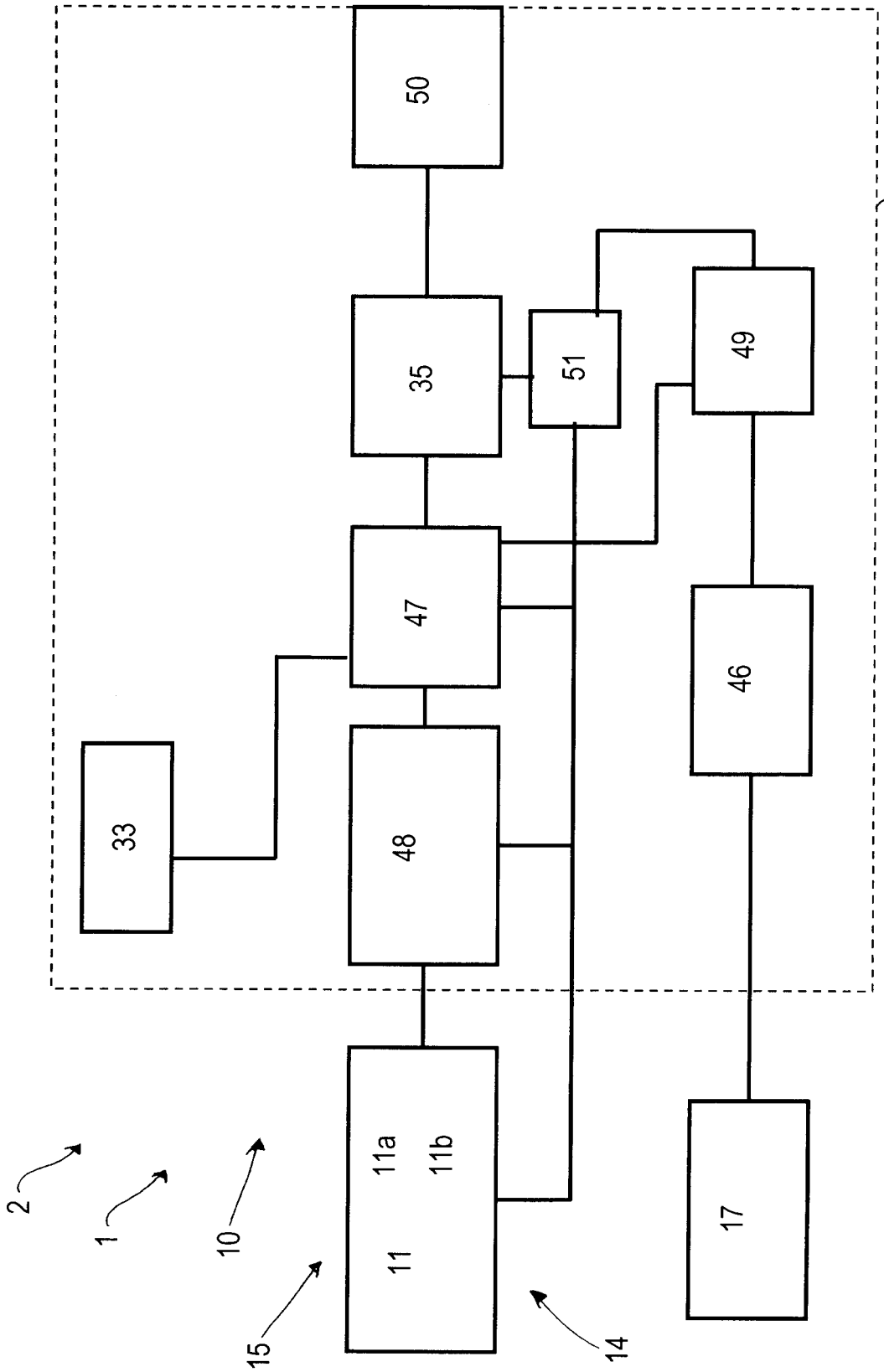


Fig. 6

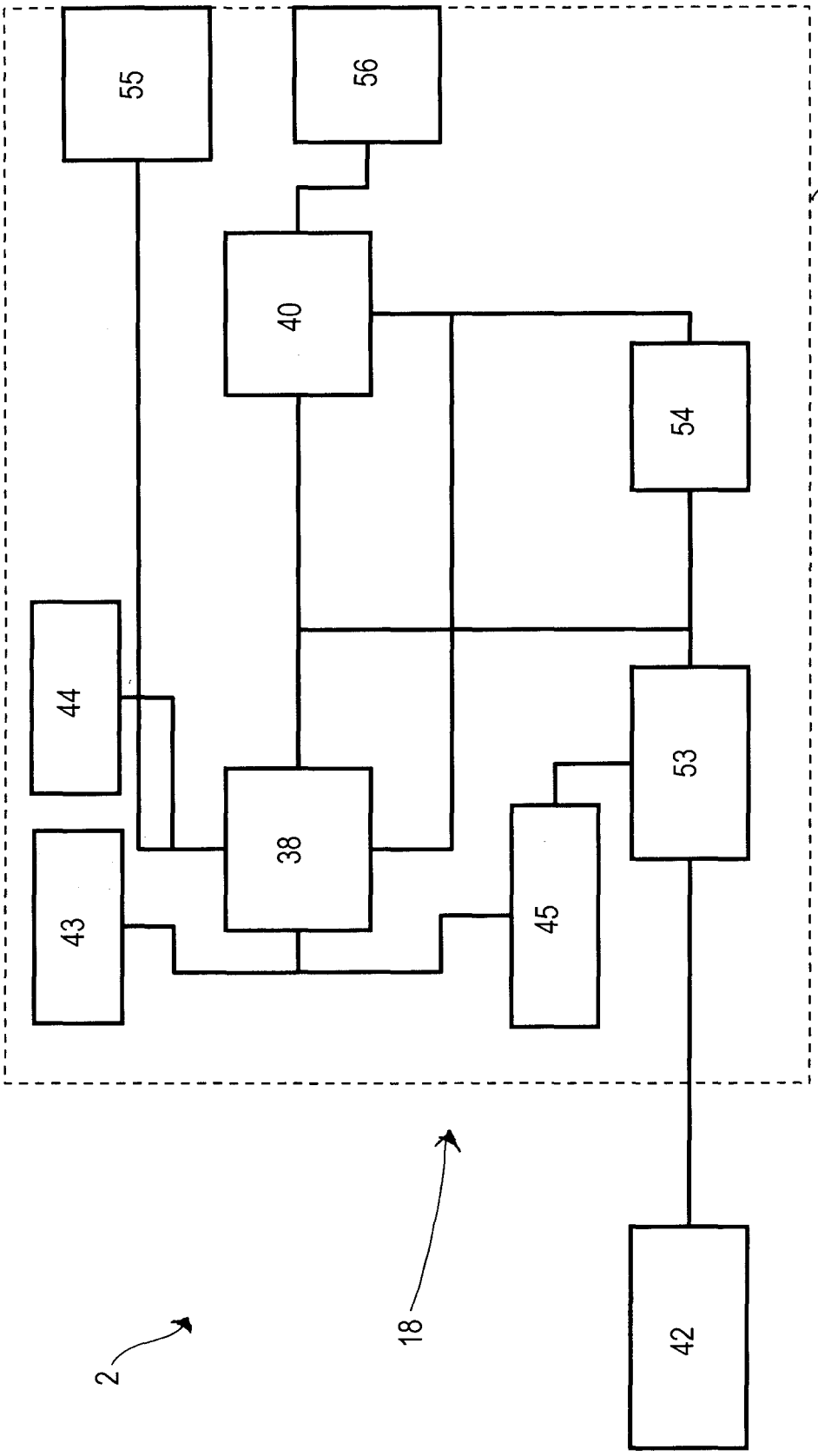


Fig. 7

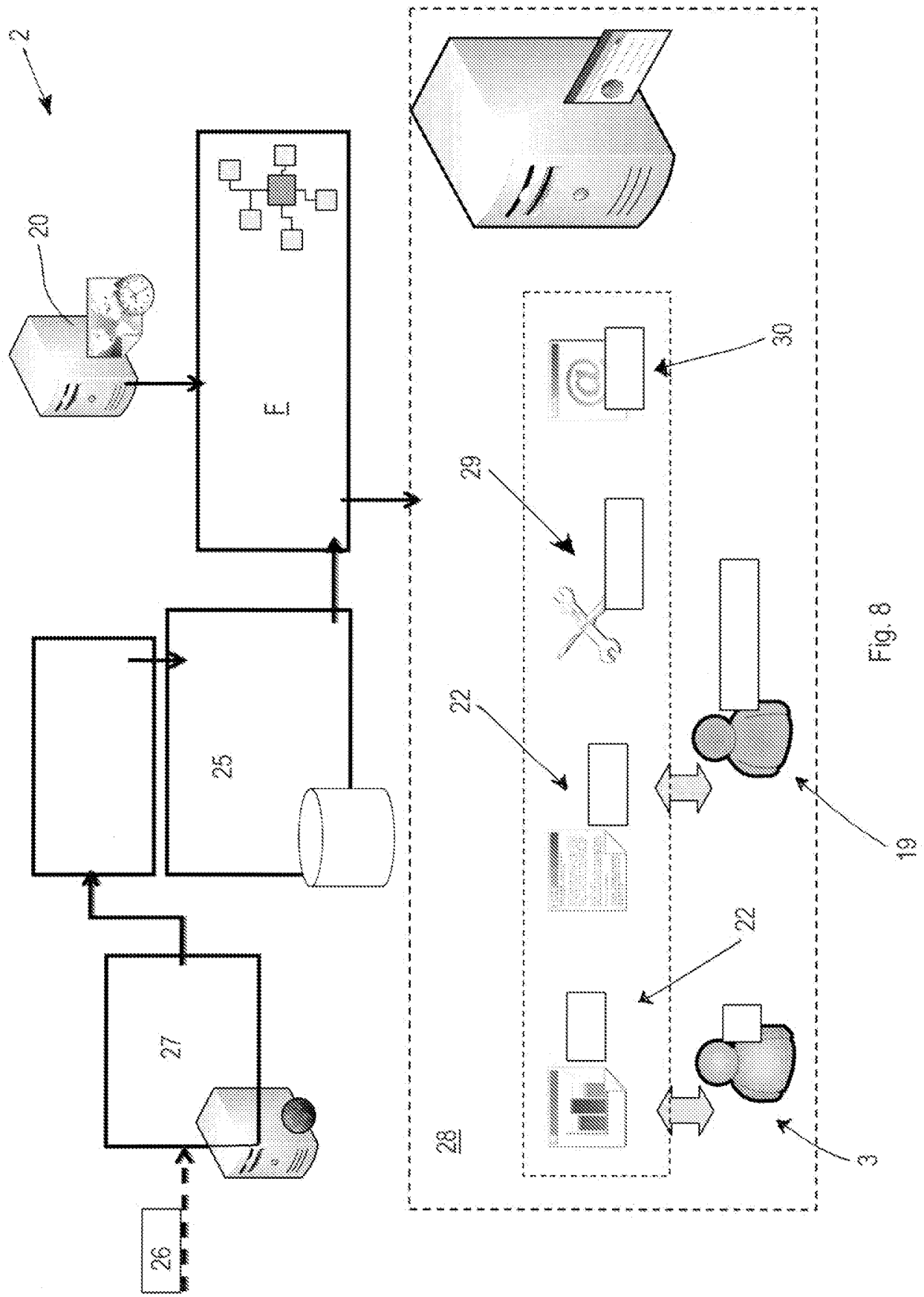


Fig. 8

REFERENCES CITED IN THE DESCRIPTION

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